Chapter 1

Introduction to Bayesian Computing Challenges

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Suppose a test is positive. How likely is the tested person to have the condition? Or suppose sales increased after an advertising campaign. How much of that increase should be attributed to advertising?

These questions have the same structure: an unobserved quantity must be inferred from an imperfect observation. A probability model describes that imperfection, and Bayesian inference updates uncertainty when observations become available.

The challenge is not merely to write down Bayes' rule. In the real world, the answer may require a sum over millions of unobserved states or an integral over hundreds of dependent parameters. Moreover, an accurate answer can be misleading if the model does not adequately describe the data.

Example 1.1 (Healthcare: the base rate matters)**.** Let $D \in \{0, 1\}$ indicate the condition and $T \in \{-, +\}$ the test result. Assume prevalence $\mathbb{P}(D = 1) = 0.01$, sensitivity $\mathbb{P}(T = + \mid D = 1) = 0.95$, and specificity $\mathbb{P}(T = - \mid D = 0) = 0.90$. These quantities refer to the population being tested; we assume they are known for this first calculation. By partitioning on disease status,

$$\mathbb{P}(T = +) = 0.95(0.01) + 0.10(0.99) = 0.1085, \tag{1.1}$$

$$\mathbb{P}(D = 1 \mid T = +) = \frac{0.95(0.01)}{0.1085} = \frac{19}{217} \approx 0.08756. \tag{1.2}$$

Thus, **sensitivity** and the **probability of disease** after a positive test are **different conditional probabilities**. In a hypothetical group of 10,000 people (Figure 1.1), the expected number of **true positives** is 95 and the expected number of **false positives** is 990. The latter dominates because the disease is rare.

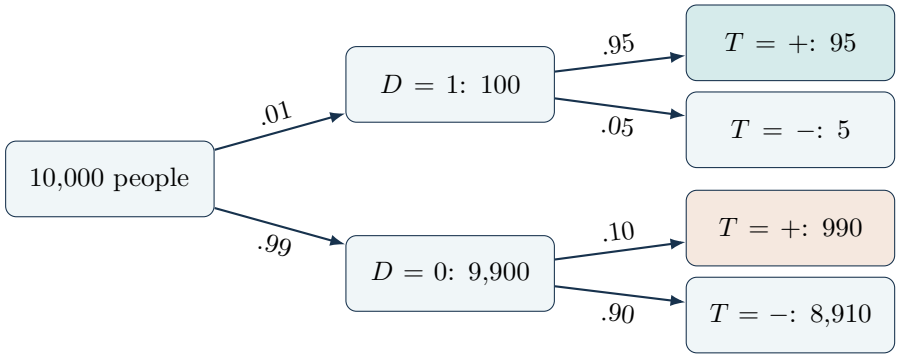


Figure 1.1: Expected counts under stipulated test characteristics. The positive branches give $95/(95 + 990)$; the diagram is an exact calculation of expectations, not a simulated patient dataset.

Here, the calculation is easy because there are only two mutually exclusive states. If **prevalence** and **test performance** must also be inferred, their uncertainties enter the integral. If symptoms, comorbidities, and repeated tests are included, the assumptions of dependence determine whether simple multiplication remains valid. For example, two test results on the same person cannot automatically be treated as independent evidence.

1.1 Learning from data

Statistical inference comes in two main flavors that we use throughout the course: **maximum likelihood** and **Bayesian** methods. In practice, each is tied to a particular kind of computation. Maximum likelihood leads to **maximization** problems: the estimator is defined implicitly as the solution of an optimization problem, whereas Bayesian inference leads to **integration** problems: the estimators are defined explicitly as integrals (Robert and Casella, 2004).

Example 1.2 (The conversion rate of a campaign)**.** An online retailer runs a promotional campaign. Out of $n = 200$ visitors who clicked on the advertisement, $s = 12$ bought something. The retailer wants to know the **conversion rate** $\theta \in (0, 1)$, that is, the probability that a visitor who clicks buys.

We encode the outcome of a visitor i as $y_i = 1$ if the visitor bought and $y_i = 0$ otherwise, and we collect the outcomes in the vector $\mathbf{y} = (y_1, \ldots, y_n)$. If visitors behave independently and each one buys with the same probability θ, then each y_i follows a Bernoulli distribution with parameter θ, written $y_i \sim \text{Bernoulli}(\theta)$:

$$p(y_i \mid \theta) = \theta^{y_i}(1 - \theta)^{1-y_i}, \qquad y_i \in \{0, 1\}.$$

Because visitors are independent, the probability of the whole observed sequence is the product of the individual probabilities:

$$p(\mathbf{y} \mid \theta) = \prod_{i=1}^{n} \theta^{y_i}(1 - \theta)^{1-y_i} = \theta^{s}(1 - \theta)^{n-s}, \qquad s = \sum_{i=1}^{n} y_i. \tag{1.3}$$

Because the data are fixed ($s = 12$, $n = 200$), (1.3) is a function of θ alone.

Figure 1.2 shows the relative likelihood function $\dfrac{L(\theta \mid y)}{\max_\theta L(\theta \mid y)}$, obtained by dividing the likelihood by its maximum value. Consequently, the curve reaches a maximum of 1 at $\hat{\theta} = \dfrac{s}{n} = \dfrac{12}{200} = 0.06.$. Values of θ close to 0.06 are therefore more compatible with the observed data. For example, $\theta = 0.05$ has relative likelihood 0.82, meaning that its likelihood is 82% of the maximum likelihood. In contrast, $\theta = 0.10$ has relative likelihood 0.13, so its likelihood is only 13% of the maximum. Thus, the data provide substantially stronger support for values of θ near 0.06 than for $\theta = 0.10$. To find the best value, we maximize the logarithm

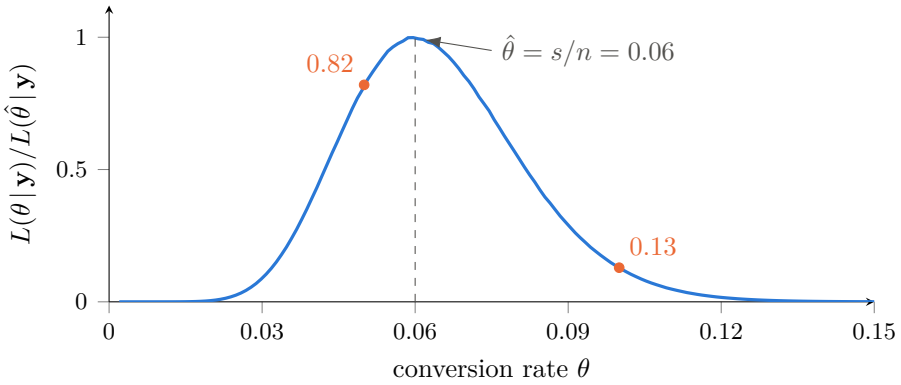


Figure 1.2: Relative likelihood $L(\theta \mid y)/L(\hat{\theta} \mid y)$ for the campaign data, where $s = 12$ buyers were observed among $n = 200$ visitors. The likelihood is maximized at $\hat{\theta} = s/n = 0.06$.

of (1.3), which is easier to differentiate and has the same maximizer because the logarithm is increasing:

$$\ell(\theta) = s \log \theta + (n - s) \log(1 - \theta), \qquad \ell'(\theta) = \frac{s}{\theta} - \frac{n - s}{1 - \theta} = 0 \iff \theta = \frac{s}{n}.$$

The best-supported conversion rate is therefore $\hat{\theta} = 12/200 = 0.06$: the observed proportion of buyers.

1.1.1 Likelihood and maximum likelihood estimation

Let's now extract the general rule from the examples.

Definition 1.1 (Likelihood)**.** Let $\mathbf{y} = (y_1, \ldots, y_n)$ be observed data and let $p(\mathbf{y} \mid \theta)$ be a statistical model: a family of probability densities (or probability mass functions) for the data, indexed by a parameter $\theta = (\theta_1, \ldots, \theta_k)$ that takes values in a parameter space Θ. The **likelihood function** is the joint density of the observed data seen as a function of the parameter, with the data held fixed:

$$L(\theta \mid \mathbf{y}) = p(\mathbf{y} \mid \theta), \qquad \theta \in \Theta.$$

When $y_1, \ldots, y_n$ are independent and identically distributed (i.i.d.) with density $p(y \mid \theta)$, the joint density factorizes and

$$L(\theta \mid \mathbf{y}) = \prod_{i=1}^{n} p(y_i \mid \theta_1, \ldots, \theta_k). \tag{1.4}$$

The **log-likelihood** is $\ell(\theta) = \log L(\theta \mid \mathbf{y})$; in the i.i.d. case, $\ell(\theta) = \sum_{i=1}^{n} \log p(y_i \mid \theta)$.

Definition 1.2 (Maximum likelihood estimator)**.** The **maximum likelihood estimator (MLE)** is the parameter value at which the likelihood attains its maximum:

$$\hat{\theta} = \arg\max_{\theta \in \Theta} L(\theta \mid \mathbf{y}) = \arg\max_{\theta \in \Theta} \ell(\theta).$$

When ℓ is differentiable and the maximum lies in the interior of Θ, $\hat{\theta}$ solves the **score equation** $\nabla_\theta \ell(\theta) = 0$, where ∇_θ is the vector of partial derivatives with respect to $\theta_1, \ldots, \theta_k$.

By construction, the MLE always lies in the parameter space Θ. Its justification is mainly asymptotic: under fairly general regularity conditions, the MLE converges almost surely to the true value of the parameter as $n \to \infty$ (Gamerman and Lopes, 2006). Two practical points are worth remembering.

- The likelihood is *not* a probability distribution over θ: it does not, in general, integrate to one in θ. Figure 1.2 is scaled to peak at one for exactly this reason.
- For n in the thousands, the product in (1.4) is too small to be stored in a computer (See Remark 1.3). We therefore always compute with $\ell(\theta)$, never with $L(\theta \mid \mathbf{y})$ directly.

1.1.2 Purchase amounts

We now apply Definition 1.2 to a model for which the answer is less obvious.

Example 1.3 (Gamma model for purchase amounts)**.** For the same campaign, the analyst records the amount spent, in euros, by $n = 10$ buyers:

$$\mathbf{y} = (32,\ 58,\ 41,\ 97,\ 25,\ 66,\ 74,\ 49,\ 120,\ 38), \qquad \sum_{i=1}^{10} y_i = 600, \qquad \bar{y} = 60.$$

The values are positive and right-skewed, so a natural model is the Gamma distribution $Y_i \overset{\text{iid}}{\sim} \text{Gamma}(\alpha, \beta)$ with density

$$p(y \mid \alpha, \beta) = \frac{1}{\Gamma(\alpha)\, \beta^{\alpha}}\, y^{\alpha - 1} e^{-y/\beta}, \qquad y > 0,$$

where $\alpha > 0$ is the *shape*, $\beta > 0$ is the *scale*, and $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t}\,\mathrm{d}t$ is the gamma function. Under this model, the expectation $\mathbb{E}[Y] = \alpha\beta$ and $\mathrm{Var}(Y) = \alpha\beta^2$. In our example, we assume that the shape is known, i.e., $\alpha = 2$, from previous campaigns, and estimate the scale parameter as β.

Log-likelihood. Substituting the density in (1.4) and using $\log \prod_i a_i = \sum_i \log a_i$,

$$\begin{aligned}\ell(\beta) &= \sum_{i=1}^{n} \left[-\log\Gamma(\alpha) - \alpha\log\beta + (\alpha-1)\log y_i - \frac{y_i}{\beta} \right] \\ &= \underbrace{-n\log\Gamma(\alpha) + (\alpha-1)\sum_{i=1}^{n}\log y_i}_{\text{does not depend on } \beta} \; - n\alpha\log\beta \; - \; \frac{1}{\beta}\sum_{i=1}^{n} y_i.\end{aligned}$$

Maximization. Setting the derivative to zero,

$$\frac{\mathrm{d}\ell}{\mathrm{d}\beta} = -\frac{n\alpha}{\beta} + \frac{1}{\beta^2}\sum_{i=1}^{n} y_i = 0 \quad \Longrightarrow \quad \hat\beta = \frac{1}{n\alpha}\sum_{i=1}^{n} y_i = \frac{\bar y}{\alpha}.$$

The critical point is a maximum: using $\sum_i y_i = n\alpha\hat\beta$,

$$\left.\frac{\mathrm{d}^2\ell}{\mathrm{d}\beta^2}\right|_{\beta=\hat\beta} = \frac{n\alpha}{\hat\beta^2} - \frac{2}{\hat\beta^3}\sum_{i=1}^{n} y_i = \frac{n\alpha}{\hat\beta^2} - \frac{2n\alpha}{\hat\beta^2} = -\frac{n\alpha}{\hat\beta^2} < 0.$$

Numbers. With $\alpha = 2$ and $\bar y = 60$, the estimate is $\hat\beta = 60/2 = 30$ euros.

Interpretation. Since $\mathbb{E}[Y] = \alpha\beta$, we have $\beta = \mathbb{E}[Y]/\alpha$; the MLE replaces the population mean $\mathbb{E}[Y]$ by the sample mean $\bar y$.

Sampling properties. Viewed as a random variable $\hat\beta = \bar Y/\alpha$ (a function of the random sample $Y_1, \ldots, Y_n$), the estimator satisfies

$$\mathbb{E}[\hat\beta] = \frac{\mathbb{E}[\bar Y]}{\alpha} = \frac{\alpha\beta}{\alpha} = \beta, \qquad \mathrm{Var}(\hat\beta) = \frac{\mathrm{Var}(\bar Y)}{\alpha^2} = \frac{\alpha\beta^2/n}{\alpha^2} = \frac{\beta^2}{n\alpha},$$

so it is unbiased. Plugging in $\hat\beta = 30$ gives an estimated standard error $\sqrt{\hat\beta^2/(n\alpha)} = 30/\sqrt{20} \approx 6.71$ euros. Its exact distribution is also known: a sum of independent Gamma variables with a common scale is Gamma, $\sum_i Y_i \sim \mathrm{Gamma}(n\alpha, \beta)$, and multiplying a Gamma variable by a constant $c > 0$ multiplies its scale by c, so

$$\hat\beta = \frac{1}{n\alpha}\sum_{i=1}^{n} Y_i \sim \mathrm{Gamma}\!\left(n\alpha, \frac{\beta}{n\alpha}\right).$$

Remark 1.1. If the shape α is *also* unknown, the score equation for α involves the derivative of $\log\Gamma(\alpha)$ and has no closed-form solution; the MLE must be found numerically. Even the "easy" flavor of inference quickly requires an algorithm. Monte Carlo methods for optimization are the subject of Chapter 4.

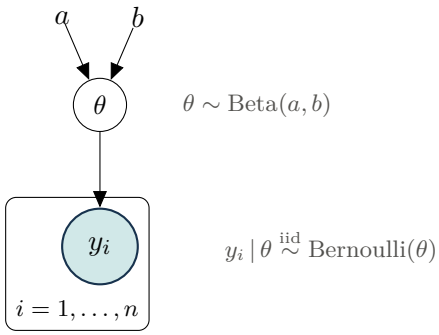


Figure 1.3: The conversion model as a Bayesian network. White circle: unknown quantity; shaded circle: observed data; the plate repeats the node for every visitor.

1.2 Bayesian inference

The MLE answers the question "which single value of θ best explains the data?". Decision makers usually ask different questions: (i) How sure are we? and (ii) What is the probability that the new campaign beats our 5% benchmark? Bayesian inference answers such questions by treating the unknown parameter as a random variable.

1.2.1 Using what we already knew

Past campaigns of the retailer converted around 5% of visitors. We express this knowledge as a probability distribution over θ before seeing the new data, the **prior**. A convenient family on $(0, 1)$ is the Beta distribution:

$$\pi(\theta) = \mathrm{Beta}(\theta \mid a, b) = \frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a,b)}, \qquad B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1}\,\mathrm{d}t = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

where $B(a, b)$ is the *Beta function*, the constant that makes the density integrate to one. A $\mathrm{Beta}(a, b)$ variable has mean $a/(a + b)$. If we choose $a = 2$ and $b = 38$, the prior mean is $2/40 = 0.05$, and it carries roughly the same weight as $a + b = 40$ past visitors.

Figure 1.3 shows the model as a Bayesian network: the hyperparameters a and b (i.e., fixed numbers) determine the distribution of θ, and θ determines the distribution of each outcome y_i. The shaded node is observed; the plate means "repeat for $i = 1, \ldots, n$".

Bayes' theorem combines prior and likelihood. Multiplying the prior by the likelihood (1.3),

$$p(\mathbf{y} \mid \theta)\,\pi(\theta) = \theta^{s}(1-\theta)^{n-s} \cdot \frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a,b)} = \frac{\theta^{a+s-1}(1-\theta)^{b+n-s-1}}{B(a,b)}.$$

As a function of θ, this has the shape of a $\mathrm{Beta}(a + s,\, b + n - s)$ density; only the constant is missing. Dividing by its integral over θ gives the **posterior**

$$\pi(\theta \mid \mathbf{y}) = \mathrm{Beta}(\theta \mid a + s,\ b + n - s) = \mathrm{Beta}(\theta \mid 14,\ 226),$$

and the integral itself, the **evidence**, is

$$p(\mathbf{y}) = \int_0^1 p(\mathbf{y} \mid \theta)\,\pi(\theta)\,\mathrm{d}\theta = \frac{B(a+s,\ b+n-s)}{B(a,b)} = \frac{B(14, 226)}{B(2, 38)} \approx 6.86 \times 10^{-21}.$$

The **evidence** is the probability that the model assigns to the exact sequence of 200 observed outcomes. It is tiny because there are 2^{200} possible sequences.

As illustrated in Figure 1.4, we can now provide a numerical answer for each question:

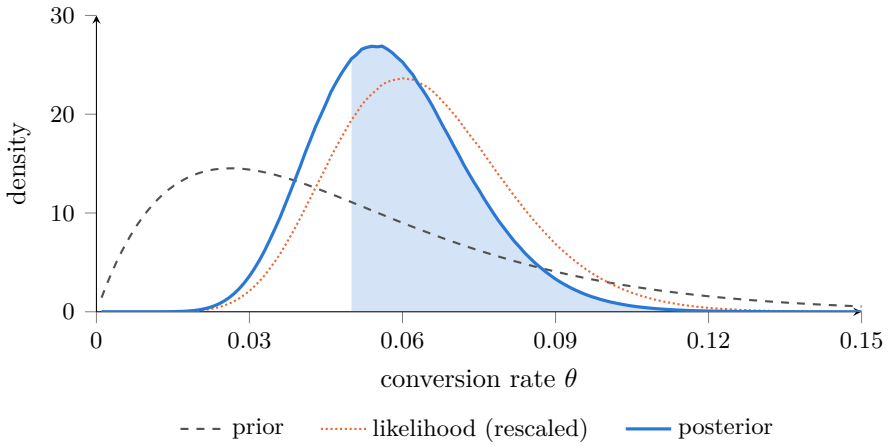


Figure 1.4: Prior Beta$(2, 38)$, likelihood (rescaled to integrate to one), and posterior Beta$(14, 226)$ for the conversion rate. The posterior is a compromise between prior and data. The shaded area is $\Pr(\theta > 0.05 \mid \mathbf{y}) \approx 0.69$.

Best guess. The posterior mean is $\mathbb{E}[\theta \mid \mathbf{y}] = \dfrac{14}{240} \approx 0.0583$, between the prior mean 0.05 and the MLE 0.06.

Uncertainty. The posterior standard deviation is 0.0151. A **95% credible interval**, an interval that contains θ with posterior probability 0.95, is $[0.032,\ 0.091]$. We can take the 2.5% and 97.5% quantiles of the posterior.

[***Decision.*** The probability that the new campaign beats the 5% benchmark is $\Pr(\theta > 0.05 \mid \mathbf{y}) \approx 0.69$.

1.2.2 Bayes' theorem and posterior summaries

Let us now state the general rule behind the example.

Definition 1.3 (Prior, posterior, evidence)**.** Let $p(\mathbf{y} \mid \theta)$ be a statistical model for data $\mathbf{y}$ with parameter $\theta \in \Theta$. A **prior** is a probability density $\pi(\theta)$ on Θ that describes what is known about θ before observing $\mathbf{y}$. Given $\mathbf{y}$, **Bayes' theorem** defines the **posterior** density

$$\pi(\theta \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \theta)\, \pi(\theta)}{p(\mathbf{y})}, \qquad p(\mathbf{y}) = \int_\Theta p(\mathbf{y} \mid \theta)\, \pi(\theta)\, \mathrm{d}\theta, \tag{1.5}$$

where $p(\mathbf{y})$ is the **evidence** (or **marginal likelihood**): the probability of the data averaged over all parameter values, weighted by the prior. It is the constant that makes the posterior integrate to one.

Because $p(\mathbf{y})$ does not depend on θ, Bayes' theorem is often written as

$$\pi(\theta \mid \mathbf{y}) \ \propto\ \tilde{\pi}(\theta \mid \mathbf{y}) = p(\mathbf{y} \mid \theta)\, \pi(\theta), \tag{1.6}$$

where $\propto$ reads "is proportional to" and $\tilde{\pi}$ denotes the **unnormalized posterior**. The unnormalized posterior is almost always easy to evaluate: it is a product of two formulas we wrote ourselves.

Definition 1.4 (Posterior expectation)**.** For a function $h : \Theta \to \mathbb{R}$, the **posterior expectation** of h is

$$\mathbb{E}_{\pi(\theta \mid \mathbf{y})}[h(\theta)] = \int_{\Theta} h(\theta)\, \pi(\theta \mid \mathbf{y})\, \mathrm{d}\theta = \frac{\int_{\Theta} h(\theta)\, p(\mathbf{y} \mid \theta)\, \pi(\theta)\, \mathrm{d}\theta}{\int_{\Theta} p(\mathbf{y} \mid \theta)\, \pi(\theta)\, \mathrm{d}\theta}. \tag{1.7}$$

Almost every Bayesian summary is of the form (1.7) for a suitable choice of h:

Summary	Choice of $h(\theta)$	Campaign example
posterior mean	$h(\theta) = \theta$	$\mathbb{E}[\theta \mid \mathbf{y}] = 0.0583$
posterior variance	$h(\theta) = (\theta - \mathbb{E}[\theta \mid \mathbf{y}])^2$	0.0151^2
probability of an event A	$h(\theta) = \mathbf{1}\{\theta \in A\}$	$\Pr(\theta > 0.05 \mid \mathbf{y}) = 0.69$
predictive probability	$h(\theta) = p(\tilde{y} = 1 \mid \theta) = \theta$	next visitor buys: 0.0583

Here $\mathbf{1}\{\cdot\}$ is the **indicator function**: it equals 1 when the condition in braces holds, and 0 otherwise, and $\tilde{y}$ denotes a future, not yet observed, outcome.

The key observation for this course is that **Bayesian inference is integration**: both the evidence and every summary are integrals over the parameter space.

The campaign example was thus easy because the posterior belonged to the same family as the prior.

Definition 1.5 (Conjugate prior)**.** A family $\mathcal{F}$ of prior distributions is **conjugate** for a model $p(\mathbf{y} \mid \theta)$ if, for every prior in $\mathcal{F}$ and every data set $\mathbf{y}$, the posterior also belongs to $\mathcal{F}$.

Proposition 1.1 (Beta–Bernoulli conjugacy)**.** *If* $\theta \sim \mathrm{Beta}(a, b)$ *and* $y_1, \dots, y_n \mid \theta \overset{iid}{\sim} \mathrm{Bernoulli}(\theta)$ *with* $s = \sum_i y_i$*, then*

$$\theta \mid \mathbf{y} \sim \mathrm{Beta}(a + s,\ b + n - s), \qquad p(\mathbf{y}) = \frac{B(a + s,\ b + n - s)}{B(a, b)}.$$

Proof. The computation of Section 1.2.1 shows $p(\mathbf{y} \mid \theta)\pi(\theta) = \theta^{a+s-1}(1-\theta)^{b+n-s-1}/B(a,b)$. Integrating over $\theta \in (0,1)$ and using the definition of the Beta function gives $p(\mathbf{y}) = B(a + s, b + n - s)/B(a,b)$. Dividing, $\pi(\theta \mid \mathbf{y}) = \theta^{a+s-1}(1-\theta)^{b+n-s-1}/B(a+s, b+n-s)$, which is the $\mathrm{Beta}(a + s, b + n - s)$ density. □

Remark 1.2 (The posterior mean as a compromise)**.** The posterior mean of Proposition 1.1 can be written as a weighted average of the prior mean and the MLE:

$$\mathbb{E}[\theta \mid \mathbf{y}] = \frac{a + s}{a + b + n} = w\, \frac{a}{a + b} + (1 - w)\, \frac{s}{n}, \qquad w = \frac{a + b}{a + b + n}.$$

In the campaign, $w = 40/240 = 1/6$ and $\mathbb{E}[\theta \mid \mathbf{y}] = \frac{1}{6}(0.05) + \frac{5}{6}(0.06) \approx 0.0583$. As n grows, $w \to 0$ and the data dominate the prior.

1.2.3 A prior on the log-odds

Example 1.4 (Non-conjugate prior)**.** Marketing teams often reason in **log-odds** rather than in probabilities, because, in a logistic regression, the effects of several channels add up on that scale.

The log-odds of the conversion rate is:

$$\eta = \text{logit}(\theta) = \log \frac{\theta}{1-\theta} \in \mathbb{R}, \qquad \text{with inverse} \qquad \theta = \sigma(\eta) = \frac{1}{1+e^{-\eta}},$$

where σ is the logistic function.

Suppose the team places a normal prior on the log-odds, $\eta \sim \mathcal{N}(\mu_0, \sigma_0^2)$, centered on the 5% benchmark, $\mu_0 = \text{logit}(0.05) \approx -2.944$, with standard deviation $\sigma_0 = 0.5$.

Writing (1.6) is still immediate:

$$\tilde{\pi}(\eta \,|\, \mathbf{y}) = \sigma(\eta)^s \left(1 - \sigma(\eta)\right)^{n-s} \frac{1}{\sqrt{2\pi}\,\sigma_0} \exp\left(-\frac{(\eta-\mu_0)^2}{2\sigma_0^2}\right).$$

But no known family has this shape, and the evidence $p(\mathbf{y}) = \displaystyle\int_{\mathbb{R}} \tilde{\pi}(\eta \,|\, \mathbf{y})\, \mathrm{d}\eta$ has no closed form.

With a single parameter, we can still integrate numerically. The **trapezoidal rule** evaluates the integrand f at m equally spaced points $\eta_1 < \cdots < \eta_m$ with spacing Δ and approximates the area under the curve by a sum of trapezoids,

$$\int_{\eta_1}^{\eta_m} f(\eta)\, \mathrm{d}\eta \approx \Delta \left[\tfrac{1}{2} f(\eta_1) + f(\eta_2) + \cdots + f(\eta_{m-1}) + \tfrac{1}{2} f(\eta_m)\right].$$

On the interval $[-6, 0]$, which contains essentially all posterior mass, we obtain:

Grid points m	$p(\mathbf{y})$	**Relative error**	$\mathbb{E}[\theta \,|\, \mathbf{y}]$
5	1.66×10^{-20}	7.4×10^{-1}	0.0474
10	9.99×10^{-21}	4.8×10^{-2}	0.0605
20	9.53×10^{-21}	8.2×10^{-6}	0.0576
50	9.53×10^{-21}	1.5×10^{-7}	0.0576

The reference value comes from adaptive quadrature (See for instance, `scipy.integrate.quad`).

Twenty evaluations of $\tilde{\pi}$ already give five correct digits, and the posterior summaries follow: $\mathbb{E}[\theta \,|\, \mathbf{y}] \approx 0.0576$ and $\Pr(\theta > 0.05 \,|\, \mathbf{y}) \approx 0.685$, close to the conjugate answers because the two priors encode similar beliefs.

Thus, we now know that losing conjugacy did not make the problem harder, as in one dimension, numerical integration is cheap and accurate. The main issue begins when the parameter has many components, when the posterior has an awkward shape, or when the likelihood itself cannot be written down. These are the three obstacles that we discuss in the Section 1.3.

1.3 Why Bayesian computation is hard

Bayesian inference begins with a disarmingly simple statement, (1.5). The numerator is usually easy to write down. The difficulty lies in the denominator, the evidence $p(\mathbf{y})$, which is an integral over the entire parameter space, and in the posterior expectations (1.7) that we ultimately care about. Three obstacles recur throughout the course:

Intractable normalizing constants. Computing $p(\mathbf{y})$ requires integrating over a possibly high-dimensional parameter space. Deterministic numerical integration (e.g., quadrature) is accurate in one or two dimensions, but its cost explodes as the number of parameters grows (Section 1.3.1).

High-dimensional and multimodal posteriors. Realistic models, such as hierarchical, latent-variable, and graphical models, live in spaces with tens, hundreds, or thousands of dimensions. There, the posterior mass concentrates on thin regions that are hard to locate, and it may be split between several separated peaks (Section 1.3.2).

Intractable likelihoods. In some models, we cannot even evaluate $p(\mathbf{y} \mid \theta)$, although we can simulate data from the model (Section 1.3.3).

1.3.1 Obstacle 1: normalizing constants and the curse of dimensionality

From a business question to a probability model. The retailer does not run a single advertisement: it spends on television, paid search, social media, and other channels, and wants to know how much of its sales each channel drives.

A **marketing mix model (MMM)** (also called a media mix model) relates aggregate sales to advertising and other drivers (Figure 1.5).

Let y_t be centered sales, in fixed units, in week $t = 1, \ldots, T$, and let $x_{tc} \geq 0$ be the advertising spend on channel $c = 1, \ldots, C$.

Let $u_t \in \mathbb{R}^J$ contain J measured controls, such as seasonality or a planned promotion. One illustrative model is

$$a_{tc} = x_{tc} + \lambda_c a_{t-1,c}, \qquad a_{0c} = 0, \quad 0 < \lambda_c < 1, \tag{1.8}$$

$$g(a; \kappa_c) = \frac{a}{\kappa_c + a}, \qquad \kappa_c > 0, \tag{1.9}$$

$$y_t \mid \theta \overset{\text{ind}}{\sim} \mathcal{N}\left(\alpha + \gamma^\top u_t + \sum_{c=1}^{C} \beta_c g(a_{tc}; \kappa_c), \sigma^2\right). \tag{1.10}$$

The recurrence represents **carryover**: the **adstock** a_{tc} accumulates past spend, which decays at rate λ_c. The function g represents **saturation**: extra adstock has a diminishing effect, and κ_c is the adstock at which channel c reaches half of its maximal effect β_c.

The **unknowns** are $\theta = (\alpha, \gamma, \beta_1, \ldots, \beta_C, \lambda_1, \ldots, \lambda_C, \kappa_1, \ldots, \kappa_C, \sigma)$, where α is the *baseline level*, $\gamma \in \mathbb{R}^J$ the *control effects*, and σ the *noise standard deviation*. Finally, "ind" in (1.10) means that the weekly sales are independent given θ.

It is important to note that in this example, the adstock is unnormalized: a steady spend of x yields a long-run adstock of $x/(1 - \lambda_c)$.

Consequently, changing λ_c therefore changes both persistence and level, which can increase dependence with κ_c and β_c.

This model illustrates the carryover and the shape effects studied by Jin et al. (2017).

Comparison with a model that ignores carryover. Setting $\lambda_c = 0$ makes $a_{tc} = x_{tc}$: current sales depend on current spend, with no advertising memory. With $a_{0c} = 0$, a three-week spend pulse $(10, 0, 0)$ instead produces adstock $(10, 5, 2.5)$ when $\lambda_c = .5$. At $\kappa_c = 1$, its transformed mean contributions are $\beta_c(10/11, 5/6, 5/7)$, whereas the no-carryover model gives $\beta_c(10/11, 0, 0)$.

Both are legitimate hypotheses, but they make different predictions about sales after spending stops. Even though the residuals in Equation (1.10) are conditionally independent, the mean retains exposure history.

To make the model Bayesian, we can choose proper priors on all unknowns, for example, independent $\lambda_c \sim \text{Beta}(2, 2)$, $\log \kappa_c \sim \mathcal{N}(0, 1)$, $\beta_c \sim \mathcal{N}^+(0, 1)$, $\alpha, \gamma_j \sim \mathcal{N}(0, 1)$, and $\sigma \sim \mathcal{N}^+(0, 1)$ after suitable scaling. $\mathcal{N}^+$ means a normal distribution restricted and renormalized to the positive half-line. These are illustrative priors, not scale-free defaults. In particular,

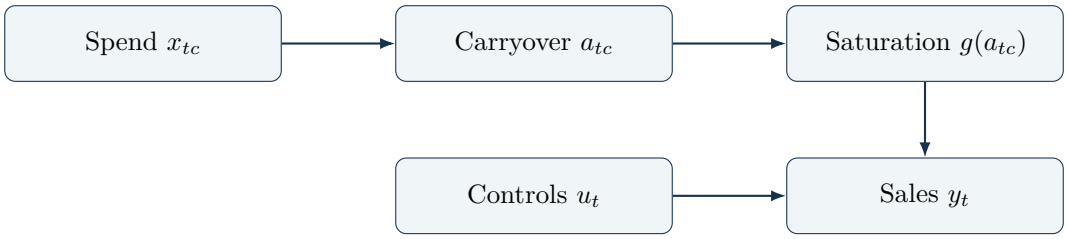


Figure 1.5: The deterministic transformations inside the mean of the MMM described in Equation (1.10). This is a calculation diagram; the observation model additionally contains noise and uncertain parameters. Initial adstock and pre-observation spend matter.

restricting β_c to be positive rules out negative channel effects. The assumption should be justified or relaxed for a real application.

How many unknowns? Each channel contributes three unknowns (β_c, λ_c, κ_c), and α, γ and σ add $J+2$ more, so θ has $d = 3C + J + 2$ components. The evidence is an d-dimensional integral,

$$p(\mathbf{y}) = \int \prod_{t=1}^{T} p(y_t \,|\, \theta)\, \pi(\theta)\, \mathrm{d}\theta,$$

and, unlike in Proposition 1.1, no conjugate structure gives it in closed form: λ_c and κ_c enter the mean nonlinearly. A modest model with $C = 3$ channels and $J = 2$ controls already has $d = 13$ unknowns.

A grid generalizes Example 1.4 by placing m points along each axis and taking all combinations, as illustrated in Figure 1.6.

It needs m^d evaluations of the unnormalized posterior. With $m = 100$ points per axis and an optimistic budget of 10^9 evaluations per second:

Unknowns d	**Evaluations** 100^d	**Computing time**
1	10^2	10^{-7} s
2	10^4	10^{-5} s
3	10^6	10^{-3} s
5	10^{10}	10 s
10	10^{20}	3.2×10^3 years
20	10^{40}	3.2×10^{23} years

The three-channel model ($d = 13$) lies beyond the $d = 10$ row: 10^{26} evaluations, or about 3×10^9 years. A model with ten channels ($d \geq 32$) is still small by industry standards and hopeless.

The phenomenon is well-known as the **curse of dimensionality**.

Definition 1.6 (Curse of dimensionality)**.** The **curse of dimensionality** refers to the exponential growth in the resources (evaluations, memory, data) needed to achieve a fixed accuracy as the dimension d increases.

For numerical integration, it can be quantified. The composite trapezoidal rule with m points per axis has an error of order m^{-2} for integrands with bounded second derivatives. With a total budget of $B = m^d$ evaluations, $m = B^{1/d}$, so

$$\text{error of a product grid} = O\Big(B^{-2/d}\Big)\,. \tag{1.11}$$

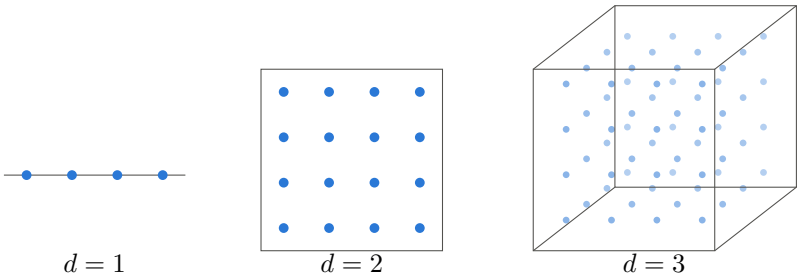


Figure 1.6: A product grid with $m = 4$ points per axis needs m^d evaluations: 4, 16 and 64 points in dimensions 1, 2 and 3.

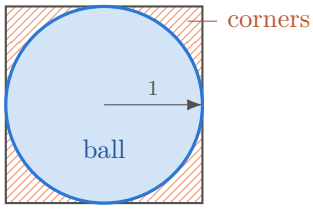


Figure 1.7: The unit ball inside the box $[-1, 1]^2$. The corners (hatched) account for 21% of the area in two dimensions and almost all of the volume in higher dimensions.

Here $O(\cdot)$ means "bounded by a constant times". For $d = 1$ the error falls like B^{-2}, but for $d = 20$ it falls like $B^{-1/10}$: multiplying the budget by 10^{10} reduces the error only tenfold. We contrast (1.11) with the Monte Carlo rate $B^{-1/2}$ discussed in Section 1.4.

Remark 1.3 (Underflow and the log-sum-exp trick)**.** A second, very practical difficulty hides in the evidence: its size. The campaign likelihood at $\theta = 0.06$ is 1.9×10^{-20} for $n = 200$ visitors, 7.2×10^{-198} for $n = 2\,000$, and e^{-4539} for $n = 20\,000$. The smallest positive number in double-precision arithmetic is about 2.2×10^{-308}, so the last value is stored as exactly 0: an **underflow**. We therefore always work with logarithms. To add numbers known only through their logarithms $v_1, \dots, v_m$ we use the **log-sum-exp** identity

$$\log \sum_{k=1}^{m} e^{v_k} = c + \log \sum_{k=1}^{m} e^{v_k - c}, \qquad c = \max_k v_k,$$

in which every exponential lies in $(0, 1]$ and at least one equals 1.

1.3.2 Obstacle 2: where the posterior mass lives

The corners of a box. Suppose we know that each channel coefficient lies in $[-1, 1]$ and that the posterior mass is concentrated near the center, within a ball of radius 1. A grid or a set of uniformly scattered points covers the whole box $[-1, 1]^d$. *What fraction of them falls inside the ball?*

In two dimensions (Figure 1.7), the disc covers $\pi/4 \approx 79\%$ of the square. In higher dimensions:

d	1	2	3	5	10	20	50
fraction in ball	1	0.79	0.52	0.16	2.5×10^{-3}	2.5×10^{-8}	1.5×10^{-28}

In twenty dimensions, only one point in forty million where the mass is. With 200 000 uniform draws, we find 0.00248 at $d = 10$ and no points at $d = 20$.

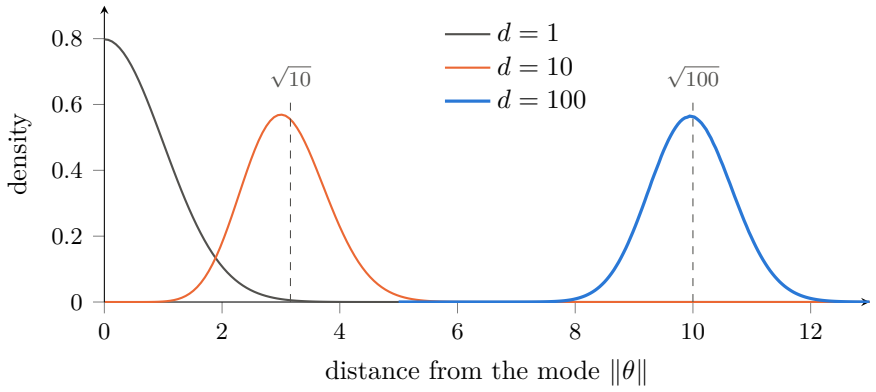


Figure 1.8: Density of the distance $\|\theta\|$ between a draw of a d-dimensional standard Gaussian and its mode. As d grows, draws concentrate in a shell of radius $\approx \sqrt{d}$.

Abstraction: concentration of measure. The table follows from the volume of the d-dimensional ball of radius r, $V_d(r) = \pi^{d/2} r^d / \Gamma(d/2+1)$, divided by the volume 2^d of the box:

$$\frac{V_d(1)}{2^d} = \frac{\pi^{d/2}}{2^d\,\Gamma(d/2+1)} \xrightarrow[d\to\infty]{} 0 \quad \text{faster than exponentially.}$$

A related and even more counterintuitive fact concerns the posterior itself. Let $\theta \sim \mathcal{N}(0, I_d)$, a standard Gaussian in d dimensions, where I_d is the $d \times d$ identity matrix. Its density is largest at the mode $\theta = 0$. Yet the squared distance $\|\theta\|^2 = \sum_{j=1}^{d} \theta_j^2$ has mean d and variance $2d$, so a random draw lies at distance about $\sqrt{d}$ from the mode, in a thin shell whose width does not grow with d (Figure 1.8). The region where most of the probability lies is called the **typical set**; in high dimensions, it is far from the mode:

d	average $\|\theta\|$	95% of draws in	$\Pr(\|\theta\| < 1)$
1	0.80	[0.03, 2.24]	0.68
10	3.08	[1.80, 4.53]	1.7×10^{-4}
100	9.98	[8.62, 11.38]	1.8×10^{-80}
1000	31.61	[30.24, 33.01]	≈ 0

Consequences: a method that explores only the neighborhood of the mode (for example, the maximum a posteriori (MAP) estimate plus a small grid around it) essentially misses all posterior mass, and a method that spreads effort uniformly wastes it in empty corners. Good methods must find the typical set and move within it; this is precisely what the Markov chains of Chapters 5 and 6 are designed to do.

Two customer segments. A second geometric trap is **multimodality**: a posterior with several separated peaks, or **modes**.

Example 1.5 (Mixture of customer segments)**.** The retailer believes that buyers come from two segments, bargain hunters and premium buyers, in equal proportions. Purchase amounts in segment $k \in \{1, 2\}$ are $\mathcal{N}(\mu_k, 10^2)$, and the unknowns are the segment means μ_1 and μ_2, each with prior $\mathcal{N}(65, 30^2)$. The likelihood of one amount is the **mixture**

$$p(y_i \mid \mu_1, \mu_2) = \tfrac{1}{2}\,\mathcal{N}(y_i;\ \mu_1, 10^2) + \tfrac{1}{2}\,\mathcal{N}(y_i;\ \mu_2, 10^2),$$

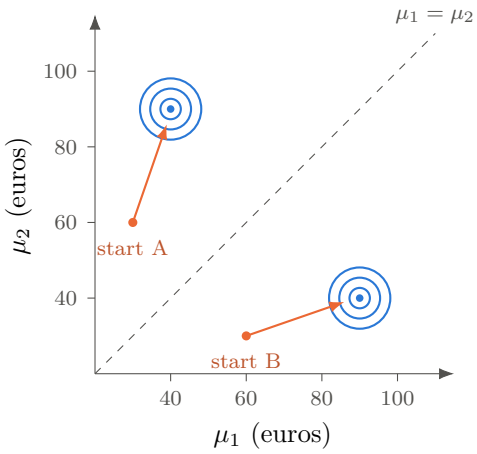


Figure 1.9: Posterior of the two segment means of Example 1.5: contours of two mirror-image modes. Gradient ascent (arrows) reaches the mode closest to its starting point.

where $\mathcal{N}(y; \mu, \sigma^2)$ denotes the normal density with mean μ and variance σ^2 evaluated at y. Swapping the labels "segment 1" and "segment 2" does not change the likelihood or the prior, so $\pi(\mu_1, \mu_2 \,|\, \mathbf{y}) = \pi(\mu_2, \mu_1 \,|\, \mathbf{y})$: every mode has a mirror image. With 100 simulated amounts (true means 40 and 90 euros), the posterior has two modes, near $(39.5, 90.0)$ and $(90.0, 39.5)$, each carrying exactly half of the mass (Figure 1.9). Gradient ascent, a local method that repeatedly steps uphill, started from $(30, 60)$ and climbs to $(39.7, 90.0)$; started from $(60, 30)$, it climbs to $(90.0, 39.7)$. Each run sees only half of the posterior, and nothing in its output reveals the existence of the other half.

With two segments, the second mode is merely a relabeling, but in general, separate modes correspond to genuinely different explanations of the data, and a sampler stuck in one of them reports overconfident answers. Detecting and handling this behavior is a central theme of Chapter 6.

1.3.3 Obstacle 3: likelihoods we can simulate but not evaluate

A referral campaign. The retailer now launches a referral program. Ten seed customers are placed in a social network of $N = 500$ people. Each day, every *active* customer persuades each neighbor who has not yet bought with probability β (the strength of word of mouth) and stops recommending with probability 0.2. The retailer observes only the daily number of new buyers $y_1, \ldots, y_T$ over $T = 40$ days.

Simulating this process is straightforward. Nevertheless, the likelihood $p(y_{1:T} \,|\, \beta)$ requires accounting for who was active on each day, which is never observed. Consequently, there are $2^N = 2^{500} \approx 10^{151}$ possible sets of active customers on a single day.

Latent variables. The referral model is an instance of a model with **latent (i.e., unobserved)** variables z, here the hidden history of the campaign. The model specifies how to generate z and then $\mathbf{y}$ given z; the likelihood is obtained by summing, or integrating, over all latent histories:

$$p(\mathbf{y} \,|\, \theta) = \int p(\mathbf{y}, z \,|\, \theta)\, \mathrm{d}z = \int p(\mathbf{y} \,|\, z, \theta)\, p(z \,|\, \theta)\, \mathrm{d}z. \tag{1.12}$$

When z is high-dimensional, (1.12) is itself an intractable integral, nested *inside* the already intractable integral (1.5).

Such a model is sometimes called a **simulator** or **implicit model**: we can run it forwards,

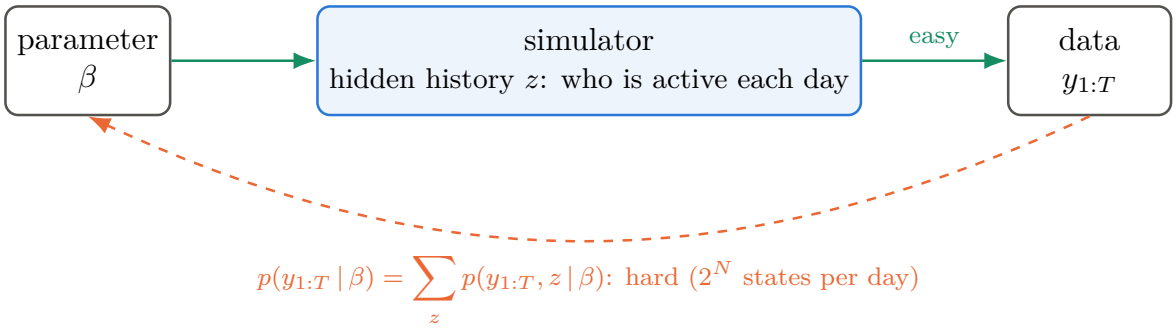


Figure 1.10: A simulator can be run forwards (from parameter to data) at low cost, but evaluating the likelihood requires summing over every hidden history z.

from θ to $\mathbf{y}$, but not backward (Figure 1.10). Epidemic models in healthcare, agent-based models of markets in finance, and customer-journey models in marketing are all of this type.

1.4 The simulation-based response: Monte Carlo

The unifying response to all three obstacles is to replace deterministic computation with stochastic simulation: instead of evaluating an integral, we draw samples and average. We first see the idea at work in the campaign, then state it in general terms.

1.4.1 Counting instead of integrating

In Section 1.2.1, $\Pr(\theta > 0.05 \mid \mathbf{y})$ was the area under the posterior density to the right of 0.05. Suppose we can draw values $\theta^{(1)}, \ldots, \theta^{(N)}$ at random from the posterior $\text{Beta}(14, 226)$; every scientific library can do this. The fraction of draws above 0.05 estimates the area, and no integral is computed:

Draws N	**Estimate of $\Pr(\theta > 0.05 \mid \mathbf{y})$**	**95% Margin of error**	**Exact value**
10	0.700	± 0.284	0.690
100	0.670	± 0.092	0.690
1 000	0.688	± 0.029	0.690
10 000	0.687	± 0.009	0.690
100 000	0.691	± 0.003	0.690

Each additional digit of accuracy costs a hundred times more draws. The margin of error in the third column is computed from the draws themselves, without knowing the exact answer (Proposition 1.2).

1.4.2 The Monte Carlo estimator

Definition 1.7 (Monte Carlo estimator)**.** Let $\theta^{(1)}, \ldots, \theta^{(N)} \overset{\text{iid}}{\sim} \pi(\theta \mid \mathbf{y})$ be N independent draws from the posterior. The **Monte Carlo (MC) estimator** of the posterior expectation (1.7) is the sample average

$$\widehat{h}_N = \frac{1}{N} \sum_{i=1}^{N} h\left(\theta^{(i)}\right). \tag{1.13}$$

In Section 1.4.1, $h(\theta) = \mathbf{1}\{\theta > 0.05\}$ and $\widehat{h}_N$ is the fraction of draws above 0.05.

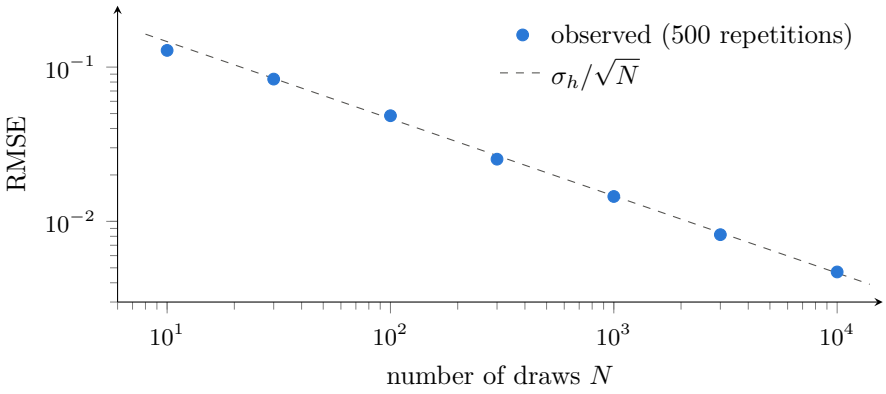


Figure 1.11: Monte Carlo error for $\Pr(\theta > 0.05 \mid \mathbf{y})$ against the number of draws, on logarithmic axes. The observed RMSE (dots, 500 repetitions) follows the prediction $\sigma_h/\sqrt{N}$ (dashed line) of Proposition 1.2.

Proposition 1.2 (Convergence and error of the Monte Carlo estimator)**.** *Let* $\mu_h = \mathbb{E}_{\pi(\theta \mid \mathbf{y})}[h(\theta)]$.

1. *(Law of large numbers (LLN).) If* $\mathbb{E}_{\pi(\theta \mid \mathbf{y})}\big[|h(\theta)|\big] < \infty$, *then* $\widehat{h}_N \to \mu_h$ *with probability one as* $N \to \infty$.

2. *(Central limit theorem (CLT).) If moreover* $\sigma_h^2 = \mathrm{Var}_{\pi(\theta \mid \mathbf{y})}\big(h(\theta)\big) < \infty$, *then* $\sqrt{N}\,(\widehat{h}_N - \mu_h)$ *converges in distribution to* $\mathcal{N}(0, \sigma_h^2)$. *In particular, the standard error of* $\widehat{h}_N$ *is* $\sigma_h/\sqrt{N}$, *and*

$$\widehat{h}_N \pm 1.96\,\frac{\widehat{\sigma}_h}{\sqrt{N}}, \qquad \widehat{\sigma}_h^2 = \frac{1}{N-1}\sum_{i=1}^{N}\big(h(\theta^{(i)}) - \widehat{h}_N\big)^2,$$

   is an approximate 95% *confidence interval for* μ_h.

Both statements are the classical limit theorems applied to the independent and identically distributed variables $h(\theta^{(1)}), \ldots, h(\theta^{(N)})$, and they are developed in Chapter 3. For an indicator h with $\mu_h = p$, the variance is $\sigma_h^2 = p(1-p)$; with $p = 0.69$, $\sigma_h \approx 0.463$. Figure 1.11 checks the $\sigma_h/\sqrt{N}$ prediction against the root-mean-square error (RMSE) observed over 500 repetitions of the experiment: for $N = 1\,000$, the observed error is 0.0145 and the prediction is 0.0146.

Remark 1.4 (The rate does not depend on the dimension)**.** Nothing in Proposition 1.2 refers to the dimension of θ: the error decreases like $N^{-1/2}$ whether θ has one component or a million. Compare with the grid rate $B^{-2/d}$ of (1.11): Monte Carlo wins as soon as $d > 4$. However, the *constant* σ_h can depend on the dimension and on how the draws are generated (Example 1.6).

Remark 1.5 (The evidence is not needed)**.** The estimator (1.13) never touches the evidence $p(\mathbf{y})$. Much of the elegance of simulation-based Bayesian inference comes from sidestepping the very integral that made (1.5) intractable. The Markov chain methods of Chapters 5 and 6 generate posterior draws using only *ratios* of the posterior at two points, in which the evidence cancels:

$$\frac{\pi(\theta' \mid \mathbf{y})}{\pi(\theta \mid \mathbf{y})} = \frac{p(\mathbf{y} \mid \theta')\,\pi(\theta')}{p(\mathbf{y} \mid \theta)\,\pi(\theta)} = \frac{\tilde{\pi}(\theta' \mid \mathbf{y})}{\tilde{\pi}(\theta \mid \mathbf{y})}.$$

Remark 1.6 (The catch)**.** Proposition 1.2 assumes that we can draw from $\pi(\theta \mid \mathbf{y})$. For the Beta posterior, this is a library call; for the posteriors of Examples 1.4 and 1.5 and of the marketing mix model, it is not. *How* to generate such draws is the subject of the rest of the course.

1.4.3 Monte Carlo against a grid

Example 1.6 (Average squared coefficient in d dimensions)**.** Take a posterior that we *can* sample, $\theta \sim \mathcal{N}(0, I_d)$, and the summary $h(\theta) = \frac{1}{d}\sum_{j=1}^{d} \theta_j^2$ (the average squared coefficient of a d-channel model). The exact answer is $\mathbb{E}[h(\theta)] = 1$ for every d. Both methods receive the same budget of $B = 10^6$ evaluations:

- *grid*: $m = \lfloor B^{1/d} \rfloor$ points per axis on $[-5, 5]$, where $\lfloor \cdot \rfloor$ rounds down, with each point weighted by the unnormalized density $e^{-\|\theta\|^2/2}$;
- *Monte Carlo*: B independent draws and the estimator (1.13).

d	**Points per axis** m	**Grid estimate**	**Monte Carlo estimate**
1	10^6	1.0000	1.0026
2	1 000	1.0000	1.0006
5	15	1.0000	0.9994
10	3	0.0853	0.9996
20	1	0.0000	1.0002

The grid is unbeatable up to $d = 5$ and useless beyond: at $d = 20$ it can afford one point per axis, placed at the mode, where $h = 0$. Monte Carlo stays within 0.003 of the truth in every dimension, in agreement with its standard error $\sqrt{2/d}/\sqrt{B} \leq 0.0014$.

The comparison is fair only because the draws came from the right distribution. If instead we estimate the normalizing constant $\int e^{-\|\theta\|^2/2}\,\mathrm{d}\theta = (2\pi)^{d/2}$ by averaging over 10^5 *uniform* draws in $[-5, 5]^d$, the ratio of estimate to truth is 1.00, 1.01, 0.99, 0.30 and 0.08 for $d = 1, 2, 5, 10, 20$: the uniform draws land in the empty corners of Figure 1.7. Choosing where to draw, through importance sampling (Chapter 3) or Markov chains (Chapters 5 and 6), is what makes Monte Carlo work in practice.

Summary

Takeaways

- **Maximum likelihood** turns inference into *optimization*, whereas **Bayesian inference** turns it into *integration* over the parameter space.
- **Bayes' theorem** (1.5) is simple to write, but the evidence $p(\mathbf{y})$ and every posterior summary (1.7) are integrals, available in closed form only for conjugate models.
- **Three obstacles make these integrals hard**: (i) the cost of grids grows like m^d, in high dimensions; (ii) the mass lies in thin, possibly disconnected regions far from the mode; and (iii) some likelihoods can only be simulated.
- Monte Carlo replaces integrals by averages of draws. Its error decreases like $\sigma_h/\sqrt{N}$ whatever the dimension, and it never needs $p(\mathbf{y})$, provided that we can generate draws from the posterior.

Further reading

The opening chapters of Robert and Casella (2004) and Gamerman and Lopes (2006) motivate simulation-based inference from the same starting point. Conjugate models and posterior summaries are covered in Chapters 2–3 of Gelman et al. (2013). The geometry of high-dimensional distributions is discussed by Bishop (2006) (Section 1.4) and Murphy (2012), and Koller and Friedman (2009) develop the Bayesian network view of Figure 1.3.

Exercises

1. *(Uniform prior.)* Repeat Section 1.2.1 with the uniform prior $\mathrm{Beta}(1, 1)$. Compute the posterior, its mean, and $\Pr(\theta > 0.05 \mid \mathbf{y})$. Explain, using the weight w of the compromise remark, why the answers change little.

2. *(Gamma–Poisson conjugacy.)* The daily number of purchases is modeled as $y_1, \dots, y_n \mid \lambda \overset{\text{iid}}{\sim} \mathrm{Poisson}(\lambda)$ with prior $\lambda \sim \mathrm{Gamma}(\alpha, \beta)$ (shape α, scale β). Show that the posterior is Gamma and give its parameters.

3. *(Unknown shape.)* In Example 1.3, let both α and β be unknown. Write the two score equations and show that the one for α involves $\psi(\alpha) = \dfrac{\mathrm{d}}{\mathrm{d}\alpha} \log \Gamma(\alpha)$. Solve them numerically for the ten purchase amounts.

4. *(Where does quadrature lose?)* Using (1.11) and Proposition 1.2, find the dimensions d for which a Monte Carlo estimate with budget B has a smaller error than a product trapezoidal rule with the same budget.

5. *(Empty balls.)* Find the smallest dimension d for which the unit ball occupies less than 1% of the box $[-1, 1]^d$. Check your answer by simulation.

6. *(Importance sampling preview.)* In Example 1.6, replace the uniform draws by draws from $\mathcal{N}(0, 2^2 I_d)$ and multiply each term by the ratio of the target to the sampling density. Compare the error with that of the uniform draws for $d = 10$ and $d = 20$.